Parent Guide — middle / module-07-eleven-multiplication

IKS Curriculum

Table of Contents

Parent Guide — Module 7: ×11 (*Ekādhikena Pūrveṇa*)

Your child's age: Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min

What your child is learning

A mental-math trick for multiplying by 11. Instead of writing out long multiplication, your child learns to:

1. **Split** the digits with a gap. (23 → 2 _ 3.)
2. **Add** the two digits. (2 + 3 = 5.)
3. **Drop** the sum in the middle. (2 5 3 = 253.)

So 23 × 11 = 253. In under 3 seconds, mentally.

The carry case (Day 2): when the digit-sum is 10 or more, the extra 1 cascades into the leading digit. Example: 87 × 11. 8 + 7 = 15. Middle digit is 5; the 1 carries into the 8 to make 9. Answer: **957**.

Why it works (Day 3): algebra. $(10a + b) \times 11 = 100a + 10(a+b) + b$. The proof is on Day 3; your child will write it themselves.

3-digit × 11 (Day 4): the pattern scales. 234 × 11 → 2 (2+3) (3+4) 4 → 2574.

×12 to ×19 (Day 6): the same idea scales proportionally — Tirthaji calls this *Anurūpyeṇa* ("proportionally"). Example: 45 × 13. Units: 5×3 = 15 (write 5, carry 1). Tens: 4×3 + 5 + 1 = 18 (write 8, carry 1). Hundreds: 4 + 1 = 5. **585.**

A note on "Vedic mathematics"

Important honest framing your child will encounter on Day 8:

- The ×11 trick (and the broader set of 16 sūtras it belongs to) was published in a book called *Vedic Mathematics* by Bharati Krishna Tirthaji in 1965, five years after his death.
- Tirthaji claimed the sūtras come from a Vedic appendix-text (*Atharva-veda Pariśiṣṭa*).

- **No other Sanskritist has produced this manuscript.** The mathematician S. G. Dani (TIFR, IIT Bombay) published a careful critique in 1993 arguing the sūtras are a 20th-century synthesis rather than a verifiably Vedic teaching.

We teach the technique enthusiastically AND we are honest about the history. The trick is real and useful; the cultural achievement (Tirthaji's pedagogical synthesis) is real; the claim of Vedic antiquity is academically disputed.

By Senior band (Grade 9–12) your child reads Dani's critique in full and forms a deeper position.

What to expect this fortnight

Daily warm-up: 2 minutes of mental ×11 chorus. Your child will do these in their head almost instantly by Day 5.

Speed comparison (Day 5): retest the Day-1 baseline problem. Most students see a 70–80% time reduction. The visible speedup is the motivation.

Project (Days 9–10): Your child picks ONE of: - A 60-second teaching video for a Grade 4 audience. - A printed workbook of 25 problems with worked examples and answer key. - A one-page proof poster deriving $(10a+b) \times 11 = 100a + 10(a+b) + b$.

How to help

Helpful: - 5-minute home drill: you call a 2-digit number, your child shouts back its ×11 product. Aim under 3 seconds per problem. - Race them: you use long multiplication, they use the trick. By Day 5 they will smoke you on speed. (Let them.) - For the proof poster: ask "Can you walk me through line 3?" — if they can verbalise the algebra, they own it.

Not helpful: - Insisting your child do the "long way" because that's how you learned. Both methods give the same answer; the trick is faster. - Asserting the technique is "ancient Vedic wisdom" without engaging with the actual evidence. Your child WILL ask you about Day 8. - Doing their project for them.

What success looks like

By the end of the module, your child can: - Compute 2-digit × 11 mentally in under 3 seconds (including carry case). - Compute 3-digit × 11 mentally in under 8 seconds. - Write out the algebraic proof: $(10a + b) \times 11 = 100a + 10(a+b) + b$. - Apply *Anurūpyeṇa* to do 2-digit × 12 through ×19 mentally. - State honestly when and by whom *Vedic Mathematics* was published, and note that the "Vedic" attribution is academically contested.

Citation standard

Projects must cite Tirthaji (1965) AND at least one of the SutraGanita PDFs (e.g. *Microsoft Word - sutras1.pdf* for the sūtra list, or *Vedic_Maths.pdf* for the practice examples). Format details in the rubric.

Want to go deeper?

Senior band (Grade 9–12) generalises the trick to polynomial multiplication and tackles the historicity debate in full (Dani 1993, Plofker 2009). Preview: `curriculum/senior/module-07-eleven-multiplication/README.md`.

Primary band (Grade 3–5) introduces the trick on no-carry 2-digit cases via a story and chant. If you have a younger sibling, look at `curriculum/primary/module-07-eleven-multiplication/`.

10-Day Lesson Plan

Module 7 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — *2 _ 3 = 253*	`days/day-01-hook.md`	Self-timed baseline; reveal
2	The Carry Case	`days/day-02-carry-case.md`	Pattern quiz
3	Why It Works	`days/day-03-why-it-works.md`	Reflection paragraph
4	Three-Digit × 11	`days/day-04-three-digit.md`	`quizzes/formative.md` Part A
5	Mental-Math Speed Day	`days/day-05-speed-day.md`	`activities/activity-01-speed-race.md` + Part B
6	*Anurūpyeṇa* — ×12 to ×19	`days/day-06-anurupyena.md`	`activities/activity-02-anurupyena.md`
7	When the Trick Doesn’t Help	`days/day-07-limits.md`	Reflection
8	Tirthaji’s Sūtra System in Context	`days/day-08-tirthaji-context.md`	Discussion
9	Mixed Practice + Project Session	`days/day-09-project-session.md`	Project draft
10	Final Assessment + Project Share	`days/day-10-final-assessment.md`	`quizzes/summative.md` + project

Pacing notes

- Day 1 establishes a personal baseline (how long to compute 73 × 11 the long way). Day 5 retests with the trick. The visible speedup is the motivation.
- Days 1–3 are the core. If your class is slow on Day 2 (the carry case), spend a second day there before moving on; combine Days 4 and 5 to recover.
- Day 6 (*Anurūpyeṇa* extension to ×12 … ×19) can be lighter or expanded depending on student appetite.
- Day 7 (“when the trick doesn’t help”) is short by design — it’s the honesty hour. ×23 doesn’t fit the family; don’t pretend it does.
- Day 8 is the cultural-context day. Keep it inquiry-driven; don’t lecture.
- Days 9–10 are project-light by design — the project here is smaller-scope than Module 5’s, because the technique is narrower.

Daily warm-up (recurring)

Every class starts with **2 minutes of “neighbour-sum” recall**: teacher calls a 2-digit number; students call back its ×11 product. Standardised set in `homework.md`.

What students leave the module with

- Confident mental ×11 on any 2-digit (and most 3-digit) number, including the carry case.
- One memorised algebraic identity: $(10a + b) \times 11 = 100a + 10(a + b) + b$.
- An extension pattern for ×12 to ×19 using *Anurūpyeṇa*.
- Honest understanding that the trick is Tirthaji’s 20th-century reformulation, not a verifiable ancient Vedic teaching.

Homework

Homework — Module 7 Eleven Multiplication (Middle)

Day 1 → Day 2

Task: Find any five 2-digit numbers (with digit-sum < 10) around your house — phone numbers, page numbers, prices. Compute each × 11 using the trick. Bring the worked list.

Time: 10 min.

Day 2 → Day 3

Task: Eight mental ×11 problems including the carry case: 76, 84, 85, 88, 93, 96, 67, 78 — each × 11. Write only the answers. Time yourself; target under 30 seconds total.

Time: 5 min.

Day 3 → Day 4

Task: Apply the proof to a number you choose. Pick any 2-digit number, write it as $10a + b$, distribute by 11, and show the final form. Show all 5 lines of working.

Time: 10 min.

Day 4 → Day 5

Task: Solve 10 three-digit × 11 problems (on the back of today's worksheet). Time yourself — goal: under 90 seconds total. Self-check three with long multiplication.

Time: 15 min.

Day 5 → Day 6

Task: 25-problem mixed drill sheet (handed out in class). Solo, untimed. Target: 100% correct.

Time: 15 min.

Day 6 → Day 7

Task: 12 problems mixing ×12, ×13, ×14 (worksheet attached). Self-check 4 with long multiplication.

Time: 15 min.

Day 7 → Day 8

Task: Find any 3 multiplication problems in your homework or a textbook. For each, decide which method (×11 family / long / other) you'd use and **why**. Write 1 sentence of justification per problem.

Time: 10 min.

Day 8 → Day 9

Task: Read the 16-sūtra list. Pick ONE sūtra (other than the two we've studied) and find what mathematical operation it's used for. Write 3 sentences. Confirm your project plan in writing (4–6 sentences).

Time: 20 min.

Day 9 → Day 10

Task: Finish your project. Polish, don't restart. Time your presentation in your head — must be 2 min.

Time: capped at 30 min.

Day 10 → end

Task: None. Module is complete. Optional: read ahead to Module 9 / 11 (Nikhilam, Ūrdhva-tiryagbhyāṁ) if curious.